Vishal Bharti

Computational Biologist · CRISPR/Cas · RNA Structural Biology · Single-Cell Genomics

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Research Profile

I build computational and statistical methods for genome engineering and RNA biology, currently in the RNA Biology Group at **CSIR-IGIB**. My own work is computational, done alongside wet-lab colleagues. The question I keep returning to is whether a biological conclusion actually follows from the data that produced it, and it runs through most of what I have done: a co-first-author method for predicting intramolecular RNA triple helices (*Biochemistry*, ACS), a single-cell analysis that identified candidate repair genes for staggered versus blunt DNA double-strand breaks, StickForStats, an open-source platform that validates the assumptions behind a statistical test before running it, and a pre-registered audit of whether published biomedical statistics can be recomputed. I came to biology from electronics engineering, taught myself the field over two years, and entered through an M.Tech at IIT Guwahati (GATE-BT **All-India Rank 127, top ~1%**).

Education

M.Tech, Biotechnology · Indian Institute of Technology (IIT) Guwahati · *2020 to 2022* · CGPA 8.75/10
Master's thesis: NMD-susceptible transcripts in UPF3B-knockout cells (RNA-seq).

B.Tech, Electronics & Communication Engineering · Institute of Engineering & Management (IEM), Kolkata · *2014 to 2018* · CGPA 8.19/10

Research Experience

Project Associate-II, RNA Biology Group (Dr. Debojyoti Chakraborty) · CSIR-Institute of Genomics and Integrative Biology (IGIB), New Delhi · **Feb 2023 to present (3+ years)**

- **RNA intramolecular triple-helix prediction (TRIPinRNA).** Co-developed (**co-first author**) a Python method to identify and characterize intramolecular RNA triple helices, applied to X-chromosome-inactivation and gene-regulation contexts; colleagues validated three of its predictions with biophysical assays. *Published in* Biochemistry (*ACS*). Co-supervised by Dr. Souvik Maiti.
- **DNA double-strand-break repair from single-cell transcriptomes.** Analyzed single-cell RNA-seq from a colleague's CRISPRi screen of DNA-repair genes, comparing cells that did and did not accurately repair a reporter cut with staggered (en1FnCas9) or blunt (SpCas9) ends. Worked from count matrices onward in Seurat (QC, integration, differential expression, gene-set comparisons, GO enrichment), producing candidate repair genes for validation; heat map and volcano plots credited in her thesis.
- **Interneuron-migration disorders in human forebrain assembloids.** Performed RNA-seq, protein-protein interaction network, and pan-tissue analyses linking engineered forebrain assembloids on polymeric nanofiber scaffolds to interneuron-migration phenotypes. (*Preprint; under revision*, Stem Cell Reports.)
- **MLC leukodystrophy: disease modeling and diagnostics.** Performed RNA-seq analysis and developed a computational approach to detect the 132dupC *MLC1* mutation, supporting iPSC disease-model work on the astrocyte-vacuolation phenotype. (*Published in* Orphanet Journal of Rare Diseases, *2026.*)
- **G-quadruplex regulation of *CCN1*.** Computational analysis and proteomics data analysis of IGF2BP1-G-quadruplex regulation of *CCN1* expression. *Published in* Biochemistry (*ACS*). In collaboration with Dr. Mary K. Ekka.

Data analysis collaborator (remote) · Archaeology PhD thesis, IIT Gandhinagar · **Mar 2025 to Jul 2025**

- Analyzed lipid residue data from archaeological pottery (GC-MS lipid profiles and compound-specific carbon isotope ratios).

- Wrote Python pipelines to clean and compile the data into one database, compute summary statistics, and produce figures for the thesis: heatmaps, box plots and isotope biplots.

Senior Research Fellow (Dr. Vibin Ramakrishnan) · IIT Guwahati · **Jun 2022 to Sep 2022**

- Proteomics, cheminformatics, and network medicine analysis of Ashwagandha and Yograj Guggulu for osteoarthritis: protein-protein interaction analysis and computational drug repurposing.

M.Tech Thesis (Dr. Kusum K. Singh) · IIT Guwahati · **2021 to May 2022**

- RNA-seq identification and functional-enrichment analysis of nonsense-mediated-decay-susceptible transcripts in UPF3B-knockout HEK293 cells.

Publications

*Name in bold; * equal contribution.*

Peer-reviewed

1. Rakheja I.*, **Bharti V.***, *et al.* (2025) "Development of an *in silico* platform (TRIPinRNA) for the identification of novel RNA intramolecular triple helices and their validation using biophysical techniques." ***Biochemistry*** (ACS) 64(1):250-265. https://doi.org/10.1021/acs.biochem.4c00334 (**Equal contribution.*)
2. Sharma S., **Bharti V.**, *et al.* (2026) "*MLC1* alteration in human iPSCs give rise to disease-like cellular vacuolation phenotype in the astrocyte lineage." ***Orphanet Journal of Rare Diseases*** 21:176. https://doi.org/10.1186/s13023-026-04316-3 (*Open access.*)
3. Rakheja I., **Bharti V.**, Singh V., Maiti S. (2026) "Quercetin affects carcinogenic phenotype of breast and lung cancer cells differentially through sterol regulation mediated by MALAT1 perturbation." ***Chemistry: An Asian Journal*** 21(11):e70821. https://doi.org/10.1002/asia.70821 (*Open access.*)
4. Das P.K., Aich M., Adu P., **Bharti V.**, Maiti S., Chakraborty D. (2026) "CRISPR-Cas diagnostics (CRISPR-Dx) of viral pathogens in low- and limited-resource areas: current state and future directions." ***TrAC Trends in Analytical Chemistry*** 195:118590. https://doi.org/10.1016/j.trac.2025.118590
5. Rana P., Ujjainiya R., **Bharti V.**, Maiti S., Ekka M.K. (2024) "IGF2BP1-mediated regulation of *CCN1* expression by specific binding to a G-quadruplex structure in its 3′ UTR." ***Biochemistry*** (ACS) 63(17):2166-2182. https://doi.org/10.1021/acs.biochem.4c00172

Preprints and manuscripts

6. **Bharti V.**, Chakraborty D. (2026) "StickForStats: automated statistical assumption validation for reproducible computational biology." *bioRxiv*. https://doi.org/10.64898/2026.06.15.732278 (*First author and corresponding; under review,* BMC Bioinformatics.)
7. **Bharti V.**, Chakraborty D. (2026) "Most biomedical articles print no statistical result checkable by recomputation: a pre-registered audit with a 20,000-article out-of-sample replication." (*Manuscript in preparation; first and corresponding author.*)
8. Rauthan R., **Bharti V.**, *et al.* "An interface of genetically engineered human forebrain assembloids and polymeric nanofiber scaffolds for multiscale profiling of interneuron-migration disorders." *Research Square* preprint. https://doi.org/10.21203/rs.3.rs-3831019/v1 (*Under revision,* Stem Cell Reports.)
9. Kochar M., Rao S., Bhattacharjee S., Goel P., Timsina H., Azhar M.K., **Bharti V.**, *et al.* "Folding on the way in: vectorial substrate recognition by human Hsp60." (*Submitted to* Cell*, 2026.*)
10. Azam T., Kumar A., Singh P., **Bharti V.**, Maiti S., Ekka M.K. "Structural and functional analysis of the Chast-NUCB1 interaction in cardiac hypertrophy." (*Revised manuscript submitted,* Journal of Biological Chemistry*, Oct 2026.*)

Research Software & Computational Methods

StickForStats: statistical-analysis platform with automatic assumption validation. First-author, open-source platform whose *Guardian* engine validates the assumptions behind every test (normality, variance homogeneity, independence, outliers) before it runs, then either reports with full confidence or cascades to the correct nonparametric alternative, with a companion module that re-checks the statistical claims in uploaded manuscripts. Built to counter the assumption violations that drive irreproducible results. *Best Poster Award, EMBO Conference 2025; first-author preprint on* bioRxiv*, under review at* BMC Bioinformatics*; public beta live at* stickforstats.com.

RNA Lab Navigator: retrieval-augmented research assistant. Keyword (BM25) retrieval over a Weaviate index of internal lab documents and the primary literature, with an LLM answering from the top passages and listing the source documents. Deployed for the lab on the institute network in July 2025.

Conferences, Talks & Posters

- Poster (selected for presentation), *Emerging Science, Technology, and Innovation Conclave (ESTIC) 2026*, Bharat Mandapam, New Delhi, 27 to 29 Oct 2026.
- **Best Poster Award**, "StickForStats: Publish Research with Confidence," **EMBO Conference 2025**, CCMB Hyderabad.
- Participant, *43rd Mahabaleshwar Seminar: Molecules, Membranes and Organelles* (TIFR), Alibaug, Maharashtra, 26 Feb to 1 Mar 2025; built and presented a simulation of organelles interacting with motor proteins under varying levels of molecular crowding in a breakout session.
- Poster, "Boundary Without Fence," *Frontiers in Genome Engineering 2023*, BITS Pilani (Goa).

Awards & Honors

- **Best Poster Award**, EMBO Conference 2025 (CCMB Hyderabad).
- **GATE Biotechnology: All-India Rank 127 (top ~1%)**, 2020.
- Graduate Assistantship, Ministry of Education, Government of India (2020 to 2022).
- GATE Electronics & Communication: top ~3 percentile, 2018.

Technical Skills

- **Languages & statistics:** Python, R, C, MATLAB; statistical modeling, experimental design, and test-selection methodology.
- **Genomics & bioinformatics:** scRNA-seq, RNA-seq, ChIP-seq; molecular docking, PyMOL; network analysis (Cytoscape, STRING, Gene Ontology).
- **ML & computational methods:** deep learning; retrieval-augmented LLM systems, vector databases (Weaviate); biomedical NLP (scispaCy, BioBERT).
- **Workflow:** Git, reproducible pipelines, Streamlit; scientific writing.

Selected Computational Projects

- **Deep learning for genome annotation.** Predicted DNA N6-methyladenine (6mA) sites in the rice genome with CNN and LSTM architectures; benchmarked predictive accuracy across models.
- **Sequence classification with RNNs.** Built an LSTM classifier to label DNA sequences by intronic, exonic, and exon-intron boundary context, improving over a CNN baseline.
- **Computational-biology collaboration (UC Berkeley).** Worked with a postdoctoral researcher at UC Berkeley to automate biological data-analysis pipelines in R and Python.

Teaching & Service

- **Teaching Assistant**, IIT Guwahati (Jun 2020 to May 2022): supported a class of 300+ students.
- **Core Organizing Team (IT lead)**, *Frontiers in Genome Engineering 2023*, BITS Pilani (Goa).
- **Mentor**, "One Week One Lab" outreach program, CSIR-IGIB (2023).
- **Chairperson**, IEEE Circuits & Systems Society Student Branch Chapter, IEM Kolkata (2016 to 2018).